Development of a Low-Cost Vibration Sensor for Structural Health Monitoring: A Case Study on the Golden Gate Bridge

Jonathan Lee
Archbishop Mitty High School
San Jose, USA
jleehops30@gmail.com

2023 IEEE MIT Undergraduate Research Technology Conference (URTC) | 979-8-3503-0965-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/URTC60662.2023.10535036

***Abstract*—Structural health monitoring (SHM) is critical for ensuring the safety of aging and deteriorating bridges. This paper proposes the development and validation of a low-cost vibration sensor for SHM of bridges. A case study involving the Golden Gate Bridge assessed the sensor's effectiveness in detecting the bridge's low-frequency vibrations. The results demonstrated promising performance, confirmed by historical data. Two experiments explored effects of material thickness and corrosion on vibration. This study highlights the sensor's cost-effectiveness and viability in contributing to reliable and efficient SHM of bridges and supports the argument for its integration into IoT sensor systems.**

Keywords—bridges, vibration, accelerometer, structural health monitoring

I. Introduction

One out of three bridges in the U.S. is in need of repair or replacement [1]. With more than 46,000 of the nation's bridges considered structurally deficient [1], it is crucial to explore and develop SHM techniques and systems that can help maintain this civil infrastructure. The consequences of structurally deficient bridges include safety risks to the public and bridge closures. For example, the collapse of the I-35 W bridge resulted in 13 deaths, 98 injuries, and substantial economic loss [2]. SHM is thus vital.

The most common form of bridge monitoring is visual inspection [3]. While visual inspection is straightforward and does not require advanced and expensive equipment, it is unreliable because it lacks quantitative methods of assessment. Therefore, other structural monitoring techniques have been researched and tested to provide more effective and reliable inspection methods. One method that has been explored in recent decades is vibration-based monitoring [4], [5]. This method is based on the idea of analyzing a bridge's natural frequency to monitor its health. A vibration-based SHM system uses installed sensors on a bridge to measure its vibrations, which can be analyzed to detect possible damage on the bridge and also provide signs of deterioration. Accelerometers are commonly used in these systems to measure vibration. According to the Advanced National Seismic System (ANSS), accelerometers are classified into three classes, namely A, B, and C, based on performance. The cost of Class A accelerometers ranges from $2000-$4000, Class B from $500-$1000, and Class C from $100-$200 [6]. These expensive costs can present financial obstacles for developing and deploying wide-scale SHM systems.

Hence, this paper proposes a low-cost vibration sensor that can be integrated into an IoT sensor system for SHM. A Micro-Electro-Mechanical-Systems (MEMS) accelerometer under $15 was paired with a microcontroller under $20. The primary contribution of this paper is demonstrating practical application of the sensor in measuring the vibrations of the Golden Gate Bridge. Laboratory experiments were also conducted to (a) verify that variation in material thickness and corrosion affect vibration and (b) evaluate the sensor's potential to monitor changes in vibration caused by these factors for real-world SHM.

II. Vibration Sensor Design

A. Materials

The materials used to develop the sensor design include the ESP32 Feather V2 microcontroller, LSM6DSO accelerometer, OpenLog data logger, 32 GB microSD card, 3.7 V and 10,050 mAh lithium-ion polymer battery, and QWIIC cables. The Arduino Software Integrated Development Environment (IDE) was used for the writing and uploading of code to the ESP32 Feather V2. Microsoft Excel was used for data analysis.

B. Hardware Development

1) Microcontroller: The ESP32 Feather V2, which is powered by the lithium-ion polymer battery, controls the activity of the system, from activating data acquisition with the LSM6DSO to commanding data storage with the OpenLog. The ESP32 Feather V2 was chosen primarily for its low cost but also for its IoT capabilities as it contains WIFI support.

2) Accelerometer: The LSM6DSO is a MEMS accelerometer that measures acceleration in three axes in units of g's, where one g is equivalent to 9.8 m/s^2. It can record both static and dynamic forces of acceleration [7]. The accelerometer has an acceleration range of ±2 g to ±16 g, acceleration noise density of 70 µg/Hz, and calculated sampling frequency of ~60 Hz [8]. The LSM6DSO contains a smart storage buffer that can hold up to 9 kB of data [8], allowing for energy-efficient data transfer, which is beneficial for IoT systems.

979-8-3503-0965-2/23/$31.00 ©2023 IEEE

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE. Downloaded on September 10,2026 at 18:10:49 UTC from IEEE Xplore. Restrictions apply.

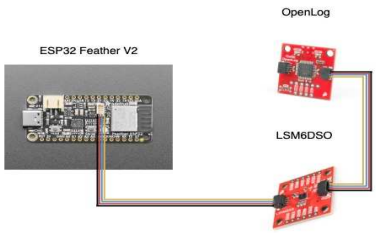


Fig. 1: Diagram of the hardware components.

C. Data Acquisition and Storage

The ESP32 Feather V2 was programmed using the Arduino IDE to read data from the accelerometer and store the data as CSV files in the microSD card contained in the OpenLog. The program records acceleration in the x, y, and z axes and time intervals of data collection.

D. Signal Processing Technique

The Fast Fourier Transform (FFT) is a signal processing algorithm that converts a time-domain signal $f(x)$ to a frequency-domain signal $F(k)$. The equation for the Fourier Transform is defined by (1) [9]. The FFT efficiently computes the Discrete Fourier Transform (DFT) of a signal and can be used to decompose a signal into its constituent frequencies.

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i k x}\, dx \qquad (1)$$

An FFT can be applied to acceleration data to determine the natural frequencies [4]. The Cooley-Tukey FFT algorithm, which requires the number of samples of the signal to be a power of 2, was implemented.

E. Calibration

The accelerometer was put through a calibration test. For each axis, the accelerometer was oriented such that the axis was parallel to the force of gravity. The acceleration was roughly 1 g for each axis, confirming the correct calibration of the accelerometer.

III. CASE STUDY: GOLDEN GATE BRIDGE

A. Background

The Golden Gate Bridge was selected for vibration measurement and analysis using the proposed vibration sensor design. The Golden Gate Bridge is a suspension bridge that is located on Highway 101 in San Francisco, California. It is constructed of steel and was completed in 1937. The objective of this case study was to demonstrate that the sensor can detect the low-frequency vibrations of the Golden Gate Bridge. The results of the natural frequencies obtained from the Golden Gate Bridge were compared to historical data to evaluate the performance of the vibration sensor.

B. Sensor Enclosure

The software Fusion 360 was used to design and 3D-print an enclosure for the vibration sensor (see Figure 2). Since the base of the enclosure was thicker than the accelerometer, a recess was optimally designed in the base to securely house the accelerometer while placing it as close as possible to the bridge. The same design step was done for the ESP32 Feather V2 and OpenLog. Another critical design consideration was to ensure stable contact between the sensor and bridge. Four holes were designed on the bottom of the base to hold magnets. The magnets were used to securely attach the sensor to the bridge. Finally, a small holder was designed to secure the battery.

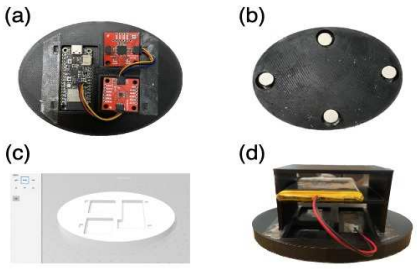


Fig. 2: (a) Vibration sensor with all three components, (b) Bottom of enclosure holding four magnets, (c) 3D-design file of base, (d) Holder with battery inside.

C. Data Collection

1) Sampling Rate: According to the Nyquist-Shannon Theorem, the accelerometer should have a sampling rate that is at least double the maximum measured frequency to prevent aliasing, a type of distortion of the measured signal [10]. The accelerometer obeyed the Nyquist-Shannon criterion as the frequency range of interest for vibrations of the Golden Gate Bridge was below 1 Hz (see Figure 5), making the accelerometer's sampling rate of ~60 Hz adequate.

2) Sensor Location: The vibration sensor was mounted on the right railing of the Golden Gate Bridge, from the reference point of the south side (San Francisco) entrance (see Figure 3). The sensor was placed at approximately the halfway point of the bridge. Data was recorded for 45 minutes.

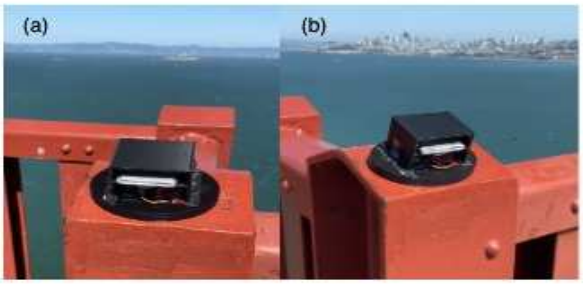


Fig. 3: (a) Front view of the vibration sensor attached to the railing of the Golden Gate Bridge at the halfway point, (b) Side view shows titled orientation of the sensor.

D. Validation

A vibration isolation test was conducted to validate the data gathered from the Golden Gate Bridge. A vibration isolation table suppresses low frequency mechanical vibrations, isolating the device under test from unwanted vibrations [11]. The

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE. Downloaded on September 10,2026 at 18:10:49 UTC from IEEE Xplore. Restrictions apply.

vibration sensor was placed on the table and acceleration data was taken for 5 minutes. (The isolation table is pictured in Figure 4.) A frequency graph corresponding to acceleration in the z-axis was produced to confirm whether the vibration data from the bridge was accurate data or simply electronic noise (see Figure 5).

IV. Vibration Response to Material Thickness

A. Background

The aim of this experiment was to verify that the thickness of a material changes the natural frequency and that the proposed vibration sensor design can detect this shift in frequency. Evaluating how the material condition of a bridge affects its vibration is critical to monitoring the structural health of a bridge. A drastic change in a bridge's natural frequency can be a sign of damage or deterioration, and one cause could be a decrease in the stiffness and strength of its material properties [4]. This could be a result of a decrease in the thickness of the bridge's structural components.

This experiment used two stainless steel strips of the same material type but different thicknesses to model cantilever beams. This difference in thickness was intentional as the thickness of a beam is related to its stiffness. Equation (2) [12] shows that a thicker beam has a larger bending stiffness.

$$k = \frac{E \times I}{L^3} \tag{2}$$

E is the Young's modulus, I is the area moment of the inertia of the beam's cross-section, and L is the length. Increasing the thickness of the beam increases its cross-sectional area and thus the area moment of inertia. Because the bending stiffness is directly proportional to the area moment of inertia, a thicker beam will have a higher bending stiffness. Moreover, a higher stiffness increases the natural frequency [13]. Therefore, it was expected that the thicker strip would have a higher natural frequency than the thinner strip.

B. Methodology

The setup of the test was designed to model a cantilever beam system. One 305 mm x 130 mm x 0.64 mm and one 305 mm x 130 mm x 0.46 mm stainless steel strip were used. It should be noted that these dimensions were obtained through direct conversion from the specifications provided by the manufacturer, which were listed in inches. The accelerometer was taped directly onto the free end of the steel strip, and the ESP32 Feather V2 and OpenLog were positioned to hang freely on the sides. The following process was completed for each strip. One end of the strip was clamped in place. (Note: The overall length of the beam was 305 mm, but because ~76 mm of the beam was clamped, the free length used for calculating the predicted natural frequency of the strip was ~229 mm.) A bending force was applied to the free end of the strip, causing it to vibrate. This was done three times to ensure the accuracy of the data. The natural frequency of the strip was obtained by performing an FFT on the data.

While there are more accurate equations to model the exact cantilever beam setup, (3) [14] was used to approximate the theoretical natural frequencies of the strips for validation of the measured data. Equation (3) is generally used to calculate the natural frequency of a mass-spring system in simple harmonic motion.

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \tag{3}$$

V. Vibration Response to Corrosion

A. Background

This experiment was conducted to verify that corrosion from rust changes the natural frequency. Corrosion is a chemical process that leads to the degradation of materials, usually metals. It is important to analyze how corrosion affects the structural integrity of a bridge since corrosion can often weaken the material of a bridge. It has been experimentally shown that steel corrosion can damage the structural health of a bridge by deteriorating the elastic modulus and thus stiffness, a process that can be observed through a reduction in the bridge's natural frequencies [15]. This test was expected to confirm that corrosion induced by rust decreases the natural frequency.

B. Methodology

A rust solution made with hydrogen peroxide, vinegar and salt was used to corrode a guitar string with a length of 711 mm (see Figure 4). The string was coated with the solution periodically throughout one day. To measure the natural frequency of the guitar string, the Spectroid app, an audio spectrum analyzer, was used instead of the vibration sensor. Spectroid was chosen as the method of vibration measurement since it can measure a much wider range of frequencies; the initial frequency of the guitar string was 331 Hz.

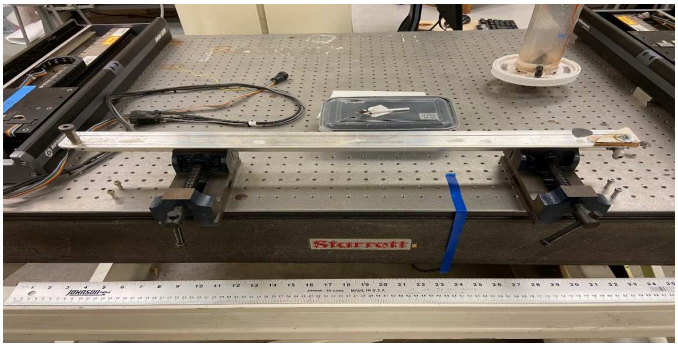


Fig. 4: Guitar string tied on both ends. Isolation table also pictured.

VI. Results and Discussion

A. Performance Assessment of Vibration Sensor on the Golden Gate Bridge

1) Vibration Analysis: The vibration sensor successfully detected low-frequency vibrations of the Golden Gate Bridge under ambient conditions (wind, traffic, etc.). The vibration response of the sensor in the z-axis during the isolation table test was plotted alongside the vibration response of the bridge in the x, y, and z axes to emphasize that the frequency peaks identified by the sensor on the bridge were real and not a result

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE. Downloaded on September 10,2026 at 18:10:49 UTC from IEEE Xplore. Restrictions apply.

of electronic noise (see Figure 5). The initial spike in amplitude exhibited by the vibration response in the x-axis was attributed to a DC offset caused by gravity, due to the sensor's titled placement on the bridge.

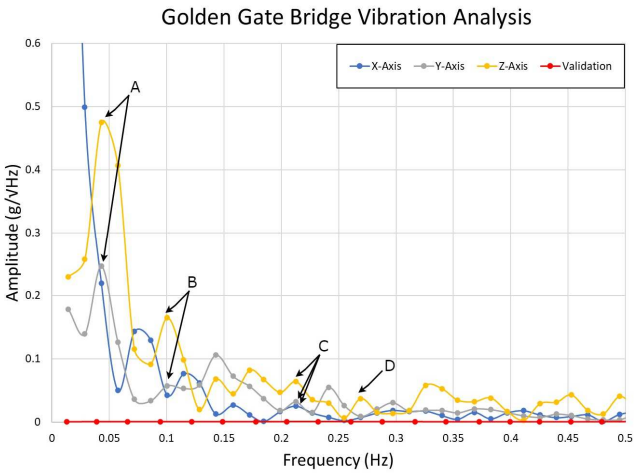


Fig. 5: Vibrational frequencies of the Golden Gate Bridge measured by the vibration sensor. The measured natural frequencies listed in Table I are identified by the labeled peaks.

2) Historical Comparison: A comparison between the data of the Golden Gate Bridge obtained by the vibration sensor and data reported in 1995 reveals remarkably similar natural frequency values from both sets, as shown in Table I. Table I summarizes the natural frequencies of the Golden Gate Bridge in four modes of vibration from 1995 [16] and presents similar natural frequencies measured by the sensor. While it cannot be definitively concluded that the measured frequencies correspond to the specific mode of vibration identified in the data from 1995, the similarities between the data validate the performance of the vibration sensor. The observed similarities also emphasize that the vibration characteristics of the Golden Gate Bridge have remained largely consistent since 1995, at least as shown by this study. These promising results indicate the potential of the vibration sensor for integration into a sensor system for SHM.

TABLE I. HISTORICAL VIBRATION DATA COMPARISON OF THE GOLDEN GATE BRIDGE

Mode Type	Natural Frequency (Hz) reported in 1995	Measured Natural Frequency (Hz) in 2023
Transverse	0.055	0.043[A]
Vertical	0.092	0.101[B]
Torsional	0.226	0.213[C]
Longitudinal	0.262	0.269[D]

An interesting observation is that the vibration response in the z-axis picked up all four frequencies listed in the third column of Table I. Moreover, it is important to note that the slight decrease observed in two of the four pairs of natural frequencies does not necessarily conclude a sign of deterioration of the bridge.

3) Limitations: There were limitations as to where the vibration sensor could be placed on the bridge due to safety reasons as well as restricted access to certain parts of the bridge. Additionally, this research used only one accelerometer, which limited further analysis of the bridge's dynamic characteristics. Factors such as wind and temperature could have also affected the readings of the accelerometer [17]. However, the results overall indicate value in the performance of the vibration sensor.

B. Effectiveness of Vibration Sensor in Monitoring Vibration Change Due to Material Thickness

The results of this experiment verified that material thickness affects the natural frequency and demonstrated specifically that a thicker material has a higher natural frequency than a thinner material. Table II shows that the thicker steel strip had a larger natural frequency than the thinner steel strip. While the difference in natural frequencies of the two strips was not large (< 1.4 Hz), there was still a decrease in the natural frequency of the thinner strip compared to the thicker strip, as predicted.

This finding was corroborated by the trend observed in the theoretical natural frequencies. Table II shows the values of the measured and theoretical natural frequencies of the two strips. The two theoretical frequencies exhibit the same trend as the measured frequencies: the higher frequency corresponds to the thicker strip, and the lower frequency corresponds to the thinner strip. It should be mentioned that the numerical values of the theoretical natural frequencies are less important than the actual trend between them.

TABLE II. MEASURED FREQUENCIES AND THEORETICAL FREQUENCIES

Strip Type	Measured Natural Frequency (Hz)	Theoretical Natural Frequency (Hz)
Thicker	5.9	5.0
Thinner	4.6	3.6

The consistency in the trend between the measured frequencies and the theoretical frequencies provides further validation of the vibration sensor's performance. The sensor's ability to capture this trend indicates its reliability in detecting subtle changes in vibration, which can be caused by variation in material strength in real bridges. The results support the viability of integrating the vibration sensor into a sensor system for SHM.

C. Vibration Sensor Potential for Monitoring Vibration Change Due to Corrosion

The guitar string exhibited a decrease in natural frequency, confirming that corrosion decreases the natural frequency (see Figure 6). While the graph does not include this final data point, the natural frequency dropped another 5 Hz to 320 Hz three days after the fourth measurement. Although the vibration sensor was not directly used in this experiment, the results support that the

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE. Downloaded on September 10,2026 at 18:10:49 UTC from IEEE Xplore. Restrictions apply.

sensor has the potential to monitor changes in vibration caused by rust and corrosion.

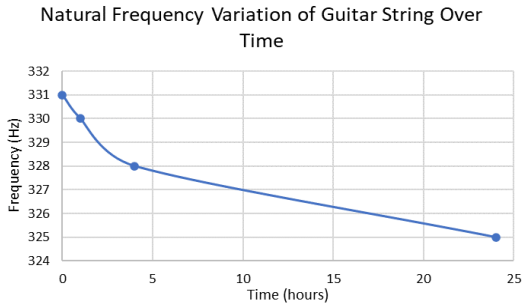


Fig. 6: Graph showing the natural frequency of the guitar string decreasing throughout 24 hours.

VII. CONCLUSION

The research showed promising results for the proposed low-cost vibration sensor. The Golden Gate Bridge case study highlighted the sensor's success in detecting low-frequency vibrations of the bridge and identifying natural frequency values. The comparison made between the measured data and the vibrations of the bridge from 1995 further validated the accuracy of the vibration sensor as well as its potential for long-term structural monitoring. The material thickness experiment demonstrated the vibration sensor's effectiveness in detecting the effect of material thickness on vibration. The corrosion experiment showcased the potential for the vibration sensor to monitor changes in the natural frequency due to rust and corrosion. Both experiments provide evidence of the sensor's potential for identifying signs of deterioration in bridges caused by these factors. Future work should explore integrating the sensor into an IoT sensor system that leverages the WIFI capabilities of the ESP32 Feather V2. Such a study would allow for more detailed analysis of a bridge's dynamic characteristics, including modal analysis. Ultimately, the proposed vibration sensor was demonstrated in this paper to be an affordable and scalable sensor option for application in SHM of bridges. Implementing such a cost-effective sensor can lead to significant cost savings for bridge maintenance agencies while still maintaining a high level of monitoring.

ACKNOWLEDGEMENTS

I would like to thank Dr. George Anwar for his advice and expertise throughout the course of this research.

REFERENCES

[1] American Road & Transportation Builders Association, "Bridge Report," artbabridgereport.org. https://artbabridgereport.org/#:~:text=1%20in%203%20U.S.%20Bridges%20Needs%20Repair%20or%20Replacement...&text=The%20length%20of%20the%20223%2C000,of%20bridges%20on%20the%20Interstate (accessed Jul. 1, 2023).

[2] S. Saidin et al., "Vibration-based approach for structural health monitoring of ultra-high-performance concrete bridge," *Case Studies in Construction Materials*. vol 18. Dec. 2020. doi: 10.1016/j.cscm.2022.e01752. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214509522008841

[3] Bridge Masters, "The Pros and Cons of Common Bridge Inspection Methods," bridgemastersinc.com. https://bridgemastersinc.com/bridge-inspection-methods/ (accessed Jul. 2, 2023).

[4] Maizur and S. Akbar. (30 Sep.-1 Oct. 2021). Estimating deterioration rate of a bridge using changes in natural frequencies. Presented at the IOP Conf. Series: Earth and Environmental Science 2nd Int. Conf. on Disaster and Management, Indonesia. [Online]. Available: https://iopscience.iop.org/article/10.1088/1755-1315/708/1/012071

[5] R. De La Torre, G. Pasobillo, M. Rebueno, D. Suñga, B. Esguerra, R. Concepcion II. (3-7 Dec. 2020). Vibration-based Structural Health Monitoring System for Bridges using ADXL345 Accelerometer with MATLAB Standalone Application. Presented at the 2020 IEEE 12th Int. Conf. on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), Manila, Philippines. [Online]. Available: https://ieeexplore.ieee.org/document/9400068

[6] A. Ali, T. Sandhu, M. Usman. "Ambient Vibration Testing of a Pedestrian Bridge Using Low-Cost Accelerometers for SHM Applications," *Smart Cities*, vol. 2, no. 1, Jan. 2019. Accessed: Jun. 14, 2023. doi: 10.3390/smartcities2010002. [Online]. Available: https://www.mdpi.com/2624-6511/2/1/2#\

[7] T. K., "Accelerometer Basics." learn.sparkfun.com. https://learn.sparkfun.com/tutorials/accelerometer-basics/all (accessed Jun. 14, 2023).

[8] STMicroelectronics, "iNEMO inertial module: always-on 3D accelerometer and 3D gyroscope," STMicroelectronics, 2019. [Online]. Available:https://cdn.sparkfun.com/assets/2/3/c/6/5/lsm6dso.pdf?_gl=1*l3lppg*_ga*Mjk5OTkyMzE0LjE2ODQxMTk0Mzc.*_ga_T369JS7J9N*MTY5MDkzMTk5OS43LjEuMTY5MDkzMjA0OS4xMC4wLjA

[9] Wolfram Alpha, "Fourier Transform." mathworld.wolfram.com. https://mathworld.wolfram.com/FourierTransform.html (accessed Jun. 3, 2023)

[10] E. Por, M. Kooten, V. Sarkovic, "Nyquist-Shannon sampling theorem," *Leiden University*, vol. 1, no. 1, pp. 1-2, May. 2019. Accessed: Jun. 20, 2023.Available:https://home.strw.leidenuniv.nl/~por/AOT2019/docs/AOT_2019_Ex13_NyquistTheorem.pdf

[11] D. Place, "Vibration Isolation Tables." blog.semiprobe.com. https://blog.semiprobe.com/vibration-isolation-table#:~:text=Vibration%20isolation%20tables%20are%20primarily,line%20or%20keep%20it%20there (accessed: Jun. 27, 2023).

[12] Amrita Vishwa Vidyapeetham, "Free Vibration of a Cantilever Beam (Continuous System)." vlab.amrita.edu. https://vlab.amrita.edu/?sub=3&brch=175&sim=1080&cnt=1 (accessed Jun. 4, 2023).

[13] Brown University, "EN4: Dynamics and Vibrations." brown.edu. https://www.brown.edu/Departments/Engineering/Courses/En4/notes_old/Freevibes/freevibes.html (accessed Jun. 7, 2023).

[14] S. Ling, "Oscillations," in *University Physics Volume 1*. Houston, TX: OpenStax, 2016, ch. 15, pp. 723-750. Accessed: Jun. 7, 2023). [Online]. Available: https://openstax.org/details/books/university-physics-volume-1

[15] Y. Zhang et al, "Natural Frequency Response Evaluation for RC Beams Affected by Steel Corrosion Using Acceleration Sensors," *Sensors (Basel)*, vol. Sep. 2020. Accessed: Jun. 16, 2023. doi: 10.3390/s20185335. [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7570730/#:~:text=Steel%20corrosion%20deteriorates%20the%20elastic,further%20reduces%20its%20natural%20frequencies

[16] A. Chopra, *Dynamics of Structures: Theory and Applications of Earthquake Engineering*. Englewoods Cliffs, NJ: Prentice Hall, 1995, pp. 35-61

[17] L. Zhang, L. Sun, L. Dong, "Experimental Study on the Relationship between the Natural Frequency and the Corrosion in Reinforced Concrete Beams," *Advances in Materials Science and Engineering*, Jul. 2021. Accessed: Jul. 2, 2023. [Online].. Available: https://doi.org/10.1155/2021/9976738

Authorized licensed use limited to: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE. Downloaded on September 10,2026 at 18:10:49 UTC from IEEE Xplore. Restrictions apply.